

# Designing a Quarter wave transformer by varying following parameters a) frequency b) input impedance c) characteristic impedance

## Test case 1:

```
clear;

clc;

Zr=100;

s=9;

d=0.1;

Zs=50;

Ro=sqrt(Zs*Zr);f=500*(10^6);

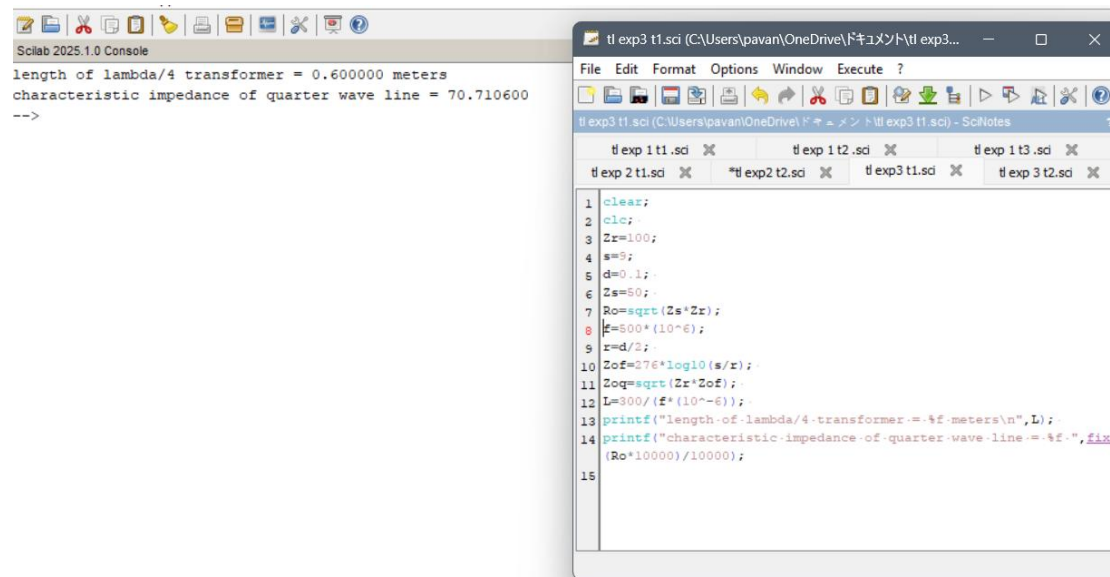
r=d/2; Zof=276*log10(s/r);

Zoq=sqrt(Zr*Zof);

L=300/(f*(10^-6));

printf("length of lambda/4 transformer = %f meters\n",L);

printf("characteristic impedance of quarter wave line = %f ",fix(Ro*10000)/10000);
```



The screenshot displays the Scilab 2025.1.0 environment. On the left, the 'Console' window shows the output of the code: 'length of lambda/4 transformer = 0.600000 meters' and 'characteristic impedance of quarter wave line = 70.710600'. On the right, the 'Editor' window shows the source code for 't1 exp3 t1.sci', which is identical to the code provided in the 'Test case 1' section. The code includes comments for each line, and the output is formatted with fixed-point notation for the characteristic impedance.

## Test case 2:

```
clear;

clc;
```

```

Zr=100;

s=9;

d=0.1;

Zs=50;

Ro=sqrt(Zs*Zr);f=100*(10^6);r=d/2;

Zof=276*log10(s/r);

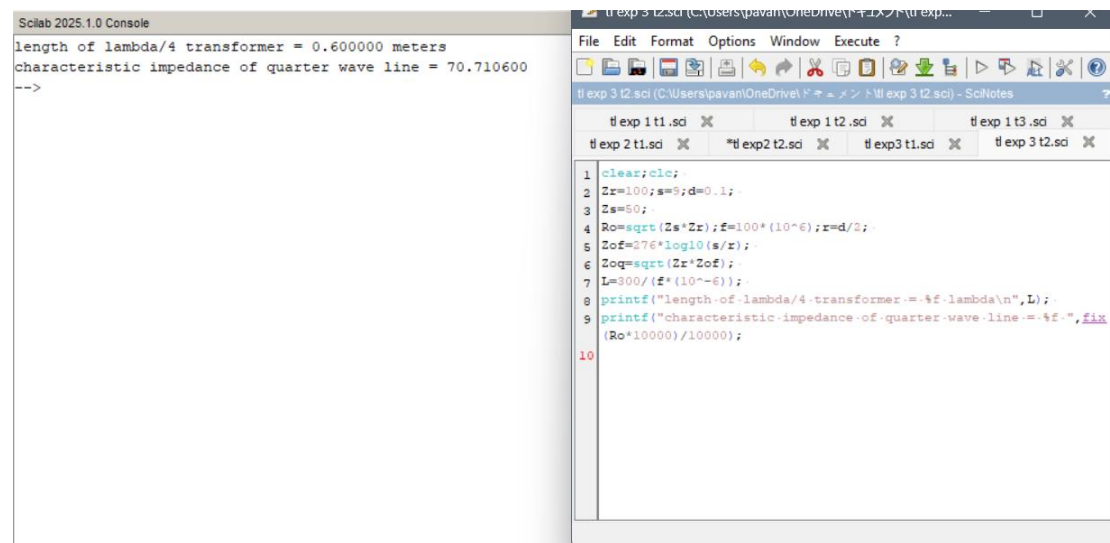
Zoq=sqrt(Zr*Zof);

L=300/(f*(10^-6));

printf("length of lambda/4 transformer = %f lambda\n",L);

printf("characteristic impedance of quarter wave line = %f ",fix(Ro*10000)/10000);

```



The screenshot displays the SciLab 2025.1.0 environment. On the left, the console window shows the output of the script: "length of lambda/4 transformer = 0.600000 meters" and "characteristic impedance of quarter wave line = 70.710600". On the right, the script editor shows the code being executed, which matches the code provided in the first block. The script includes variable assignments, calculations for characteristic impedance and length, and two printf statements for output.

```

1 clear;clc;
2 Zr=100;s=9;d=0.1;
3 Zs=50;
4 Ro=sqrt(Zs*Zr);f=100*(10^6);r=d/2;
5 Zof=276*log10(s/r);
6 Zoq=sqrt(Zr*Zof);
7 L=300/(f*(10^-6));
8 printf("length of lambda/4 transformer = %f lambda\n",L);
9 printf("characteristic impedance of quarter wave line = %f ",fix
10 (Ro*10000)/10000);

```